EEE 452 - Antenna Engineering - HW1

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Question 1: We are asked to find the pattern of an antenna whose current distribution is given as:

$$I_e(z') = \hat{a}_z I_0 \sin\left[k\left(\frac{\ell}{2} - |z'|\right)\right], z' \in \left[-\frac{l}{2}, \frac{l}{2}\right]$$

We write this as the sum of an infinite number of Hertzian dipoles. The fields due to a z-directed Hertzian dipole are given as:

$$H_\phi = j\frac{kI_e\ell\sin\theta}{4\pi r}e^{-jkr}, \quad E_\theta = j\eta\frac{kI_e\ell\sin\theta}{4\pi r}e^{-jkr}, \qquad \text{where } \ell \Rightarrow dz', I_e \Rightarrow I_e(\bar{r}), r \Rightarrow R$$

$$dH_\phi = j\frac{kI_e(z')\sin\theta}{4\pi}\frac{e^{-jkR}}{R}dz', \quad dE_\theta = j\eta\frac{kI_e(z')\sin\theta}{4\pi}\frac{e^{-jkR}}{R}dz'$$

We can calculate the E-fields as:

$$E_\theta = \int_{-\ell/2}^{\ell/2} dE_\theta = j\eta k I_0 \frac{e^{-jkr}}{4\pi r}\sin\theta \int_{-\ell/2}^{\ell/2} \sin\left[k\left(\frac{\ell}{2} - |z'|\right)\right] e^{jkz'\cos\theta} dz'$$

Notice that the integral is in the same form as PS1, Q2-a. We plug in the solution and obtain the expression for the E-field as:

$$E_\theta = j\eta I_0 \frac{e^{-jkr}}{2\pi r}\left[\frac{\cos\left(\frac{kl}{2}\cos\theta\right) - \cos\left(\frac{kl}{2}\right)}{\sin\theta}\right]$$

By using the plane wave approximation, we can obtain the H-field expression as:

$$H_\phi = jI_0 \frac{e^{jkr}}{2\pi r}\left[\frac{\cos\left(\frac{kl}{2}\cos\theta\right) - \cos\left(\frac{kl}{2}\right)}{\sin\theta}\right]$$

From these, we can obtain the field amplitude plots for the values $l = 0.25\lambda,\ 0.5\lambda,\ \lambda$ of antenna length as:

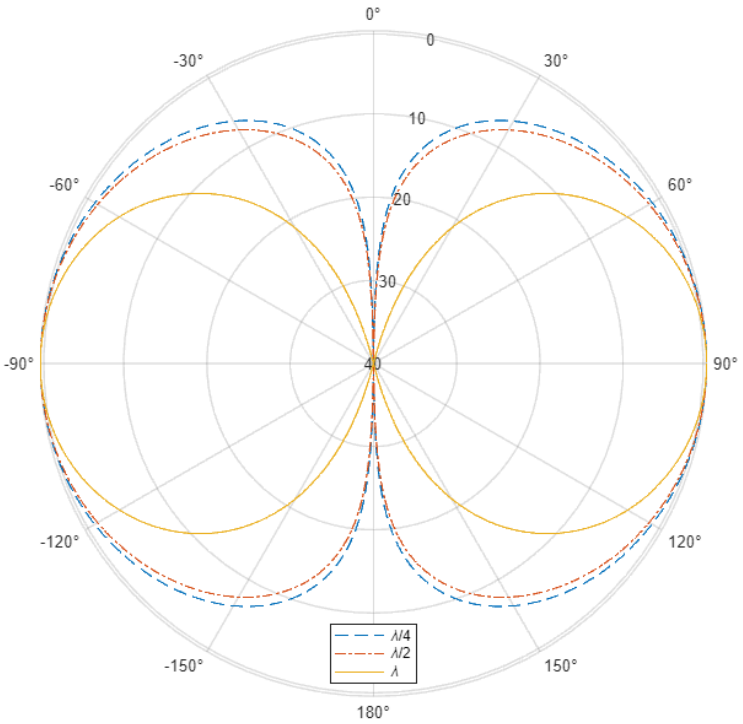

Figure 1: Field Amplitude Patterns for Linear Antennas with Various Lengths

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Question 2: Similarly, by plugging in the required value to the same expression we found in question 1, we can obtain the following plot:

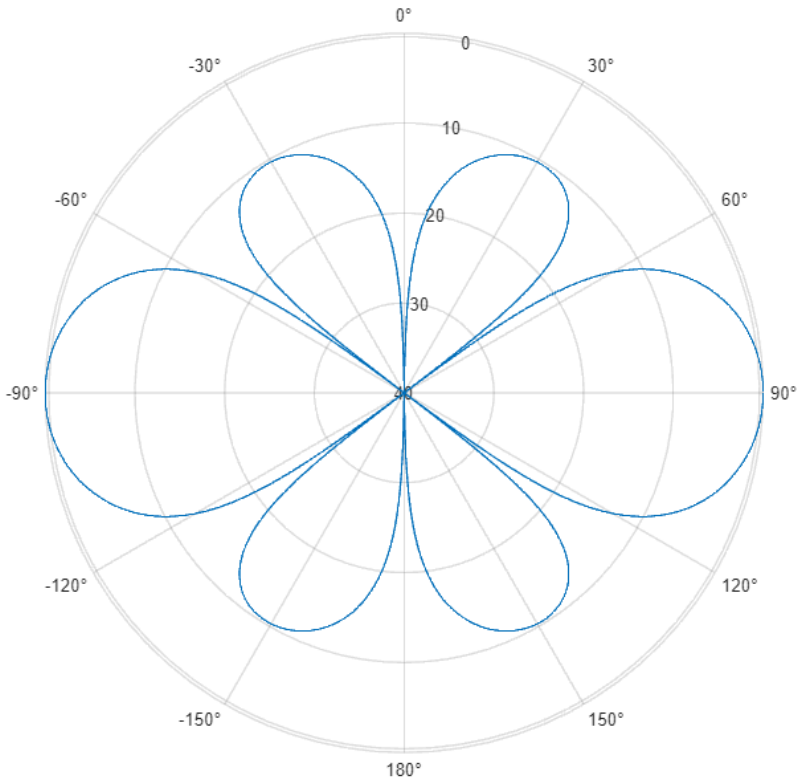

Figure 2: Field Amplitude Patterns for Linear Antenna with $l = 1.25\lambda$

Question 3: In this question, we are given a Hertzian dipole and a perfect electric conductor (PEC) plane. We can consider this geometry as two Hertzian dipoles using image theory. By modifying the expression for the fields of a Hertzian dipole, we can obtain:

$$E_\theta = j\eta \frac{kI_e\ell \sin\theta}{4\pi r} e^{-jkr}, \qquad \text{where } r \Rightarrow R = r \pm h\hat{a}_z$$

$$E_{\theta i} = j\eta k I_e \ell \sin\theta \frac{e^{-jkr}}{4\pi r} e^{\pm jkh\cos\theta}$$

If we add these two components, we obtain:

$$E_\theta = j\eta k I_e \ell \frac{e^{-jkr}}{4\pi r} \sin\theta \left[e^{jkh\cos\theta} + e^{-jkh\cos\theta}\right] = j\eta k I_e \ell \frac{e^{-jkr}}{2\pi r} \sin\theta \cos\left(kh\cos\theta\right)$$

It should be noted that this result holds for $z > 0$ or equivalently $\theta < \frac{\pi}{2}$ since, within the PEC plane, there can be no electric fields. We can go ahead and plug in $h = 0.25\lambda,\ 0.5\lambda,\ \lambda$ and plot our results.

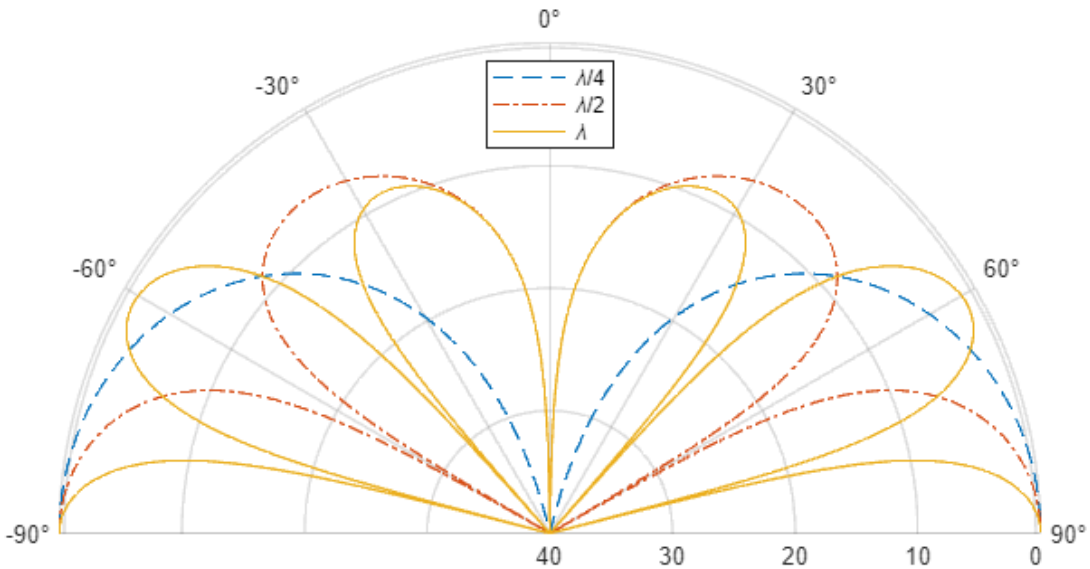

Figure 3: Field Patterns for Z-Directed Dipole above PEC with Various h Values

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Question 4: Similarly, we can plug in $h = 5\lambda$ in the expression that we found, and we can plot the result as seen in Figure 4. It should again be noted that this result holds for $z > 0$ or $\theta < \frac{\pi}{2}$.

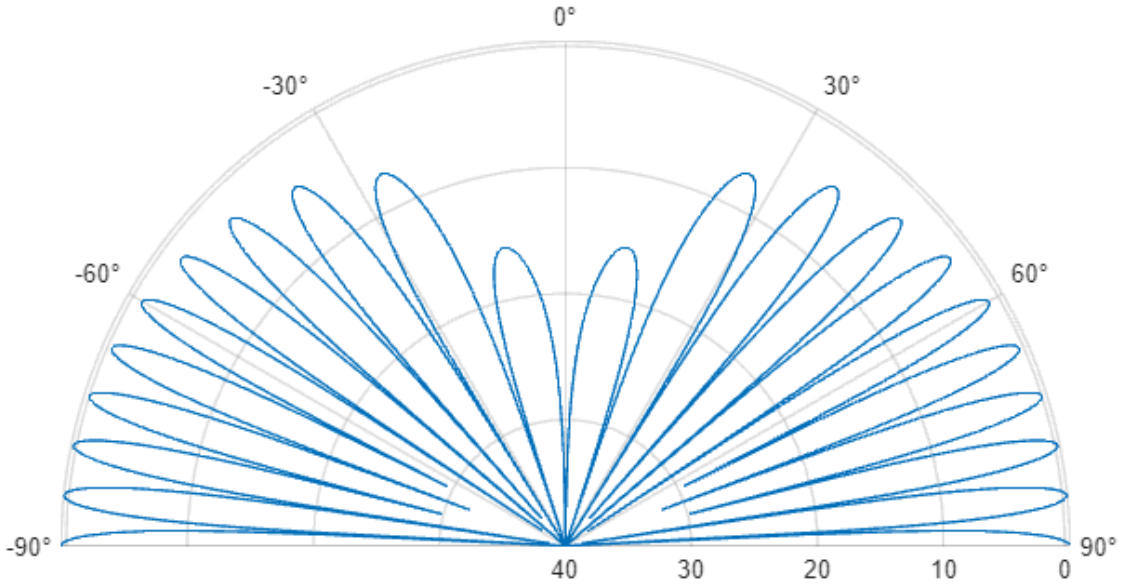

Figure 4: Field Patterns for Z-Directed Dipole above PEC with $h = 5\lambda$

Question 5: We can write the expression for a y-directed Hertzian dipole as obtained in PS5, Q2-d. We can also model this geometry as two y-directed Hertzian dipoles carrying current in opposite directions, using image theory.

$$E_\theta = -jI_e\ell\frac{k^2}{\omega\epsilon}\cos\theta\sin\phi\frac{e^{-jkr}}{4\pi r}, \quad E_\phi = -jI_e\ell\frac{k^2}{\omega\epsilon}\cos\phi\frac{e^{-jkr}}{4\pi r}, \quad \text{where } r \Rightarrow R_i = r \pm h\hat{a}_z, I_e = \pm I_0$$

We can add the fields due to both dipoles and find the fields as:

$$E_\theta = 2I_0\ell\frac{k^2}{\omega\epsilon}\frac{e^{-jkr}}{4\pi r}\cos\theta\sin\phi\left[\frac{e^{jkh\cos\theta} - e^{-jkh\cos\theta}}{2j}\right] = 2I_0\ell\omega\mu\frac{e^{-jkr}}{4\pi r}\cos(\theta)\sin(\phi)\sin(kh\cos\theta)$$

$$E_\phi = 2I_0\ell\frac{k^2}{\omega\epsilon}\frac{e^{-jkr}}{4\pi r}\cos\phi\left[\frac{e^{jkh\cos\theta} - e^{-jkh\cos\theta}}{2j}\right] = 2I_0\ell k\mu\frac{e^{-jkr}}{4\pi r}\cos(\phi)\sin(kh\cos\theta)$$

For $\phi = \frac{\pi}{2}$, these expressions simplify further:

$$E_\phi = 0, E_\theta = 2I_0\ell\omega\mu\frac{e^{-jkr}}{4\pi r}\cos(\theta)\sin(kh\cos\theta)$$

Again, due to the PEC plane at $z = 0$, these expressions only hold for $z > 0$ or $\theta < \frac{\pi}{2}$. However, we have an entirely different case for $h \to 0$. We need to calculate the limit as:

$$U = r^2\frac{|E_0|^2}{2\eta}, U_{normalised} = \cos^2\theta\frac{\sin^2(kh\cos\theta)}{\sin^2(kh)}$$

$$\lim_{h\to 0} U_{normalised} = \cos^2\theta cos^2\theta, \qquad \phi = \frac{\pi}{2}$$

This expression is obtained for the power pattern, for the field pattern take the square root:

$$U = \cos^4\theta, \quad E \propto cos^2\theta$$

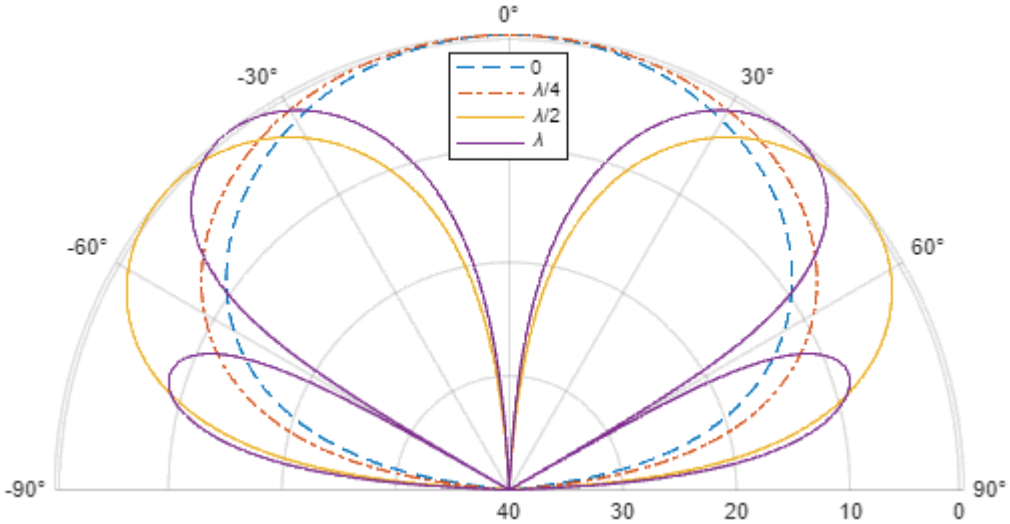

Figure 5: Field Patterns for Y-Directed Dipole above PEC with Various h Values

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Question 6: Similarly we can obtain the solution for $h = 5\lambda$ by plugging it in. Again, due to PEC plane at $z = 0$, these expressions only hold for $z > 0$ or $\theta < \frac{\pi}{2}$.

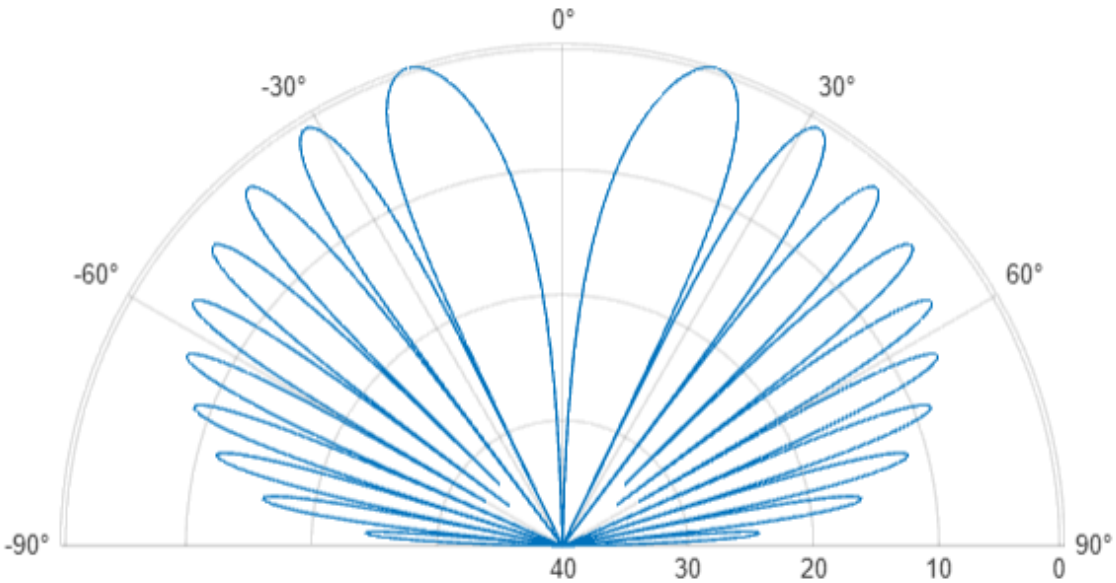

Figure 6: Field Patterns for Y-Directed Dipole above PEC with $h = 5\lambda$

Question 7: This question features a loop antenna. We can find the fields of a loop antenna as follows:

$$\bar{J}(\bar{r}) = I_0\delta(r'-a)\delta(\theta-\frac{\pi}{2})\hat{a}_\phi$$
$$\bar{A}(\bar{r}) = \frac{\mu}{4\pi}\frac{e^-jkr}{r}I_0\int \hat{a}'_\phi e^{jka(\hat{a}'_\rho\cdot\hat{a}_r)}r'd\phi'|_{r'=a,\theta'=\frac{\pi}{2}}$$

We can substitute the values as:

$$\hat{a}'_\phi = -\sin\phi'\hat{a}_x + \cos\phi'\hat{a}_y, \quad \hat{a}_r = \sin\theta\cos\phi\hat{a}_x + \sin\theta\sin\phi\hat{a}_y + \cos\theta\hat{a}_z,$$
$$\hat{a}'_\rho = \cos\phi'\hat{a}_x + \cos\phi'\hat{a}_y$$

When we use expressions, we obtain:

$$\begin{aligned}
A_r(\bar{r}) &= a\frac{\mu}{4\pi}\frac{e^{-jkr}}{r}I_0\int_0^{2\pi}\left[\cos\phi'\sin\theta\sin\phi - \sin\phi'\cos\theta\cos\phi\right]e^{jka\sin\theta\cos(\phi-\phi')}d\phi' \\
&= a\frac{\mu}{4\pi}\frac{e^{-jkr}}{r}I_0\sin\theta\int_0^{2\pi}\sin(\phi-\phi')e^{jka\sin\theta\cos(\phi-\phi')}d\phi' \qquad . \\
A_\theta(\bar{r}) &= a\frac{\mu}{4\pi}\frac{e^{-jkr}}{r}I_0\int_0^{2\pi}\left[\cos\phi'\cos\theta\sin\phi - \sin\phi'\cos\theta\cos\phi\right]e^{jka\sin\theta\cos(\phi-\phi')}d\phi' \\
&= a\frac{\mu}{4\pi}\frac{e^{-jkr}}{r}I_0\cos\theta\int_0^{2\pi}\sin(\phi-\phi')e^{jka\sin\theta\cos(\phi-\phi')}d\phi' \\
A_\phi(\bar{r}) &= a\frac{\mu}{4\pi}\frac{e^{-jkr}}{r}I_0\int_0^{2\pi}\left[\cos\phi\cos\phi' + \sin\phi\sin\phi'\right]e^{jka\sin\theta\cos(\phi-\phi')}d\phi' \\
&= a\frac{\mu}{4\pi}\frac{e^{-jkr}}{r}I_0\int_0^{2\pi}\cos(\phi-\phi')e^{jka\sin\theta\cos(\phi-\phi')}d\phi'
\end{aligned}$$

We can calculate the integrals as:

$$\int_0^{2\pi}\cos(\phi-\phi')e^{jka\sin\theta\cos(\phi-\phi')}d\phi'$$

We can seperate this into two integrals as:

$$\int_0^{\pi}\cos(\phi-\phi')e^{jka\sin\theta\cos(\phi-\phi')}d\phi' + \int_\pi^{2\pi}\cos(\phi-\phi')e^{jka\sin\theta\cos(\phi-\phi')}d\phi'$$

Shift the second one by one period to the left and change signs to obtain:

$$\int_0^{\pi}\cos(\phi-\phi')e^{jka\sin\theta\cos(\phi-\phi')}d\phi' + \int_0^{\pi}\cos(\phi+\phi')e^{jka\sin\theta\cos(\phi+\phi')}d\phi' = 2j\pi J_1(ka\sin\theta)$$

We know that both:

$$\int_0^{2\pi}\cos(\phi-\phi')e^{jka\sin\theta\cos(\phi-\phi')}d\phi' = 2j\pi J_1(ka\sin\theta)$$
$$\int_0^{2\pi}\left[cos(\phi-\phi') + j\sin(\phi-\phi')\right]e^{jka\sin\theta\cos(\phi-\phi')}d\phi' = 2j\pi J_1(ka\sin\theta)$$

Hence

$$\int_0^{2\pi}\sin(\phi-\phi')e^{jka\sin\theta\cos(\phi-\phi')}d\phi' = 0$$

We can use these integral results and obtain:

$$A_r = A_\theta = 0, \quad A_\phi = a\frac{\mu}{4\pi}\frac{e^{-jkr}}{r}I_0 2j\pi J_1(ka\sin\theta), \quad E_\phi = -j\omega A_\phi = a\frac{\mu\omega}{2}\frac{e^{-jkr}}{r}I_0 J_1(ka\sin\theta)$$

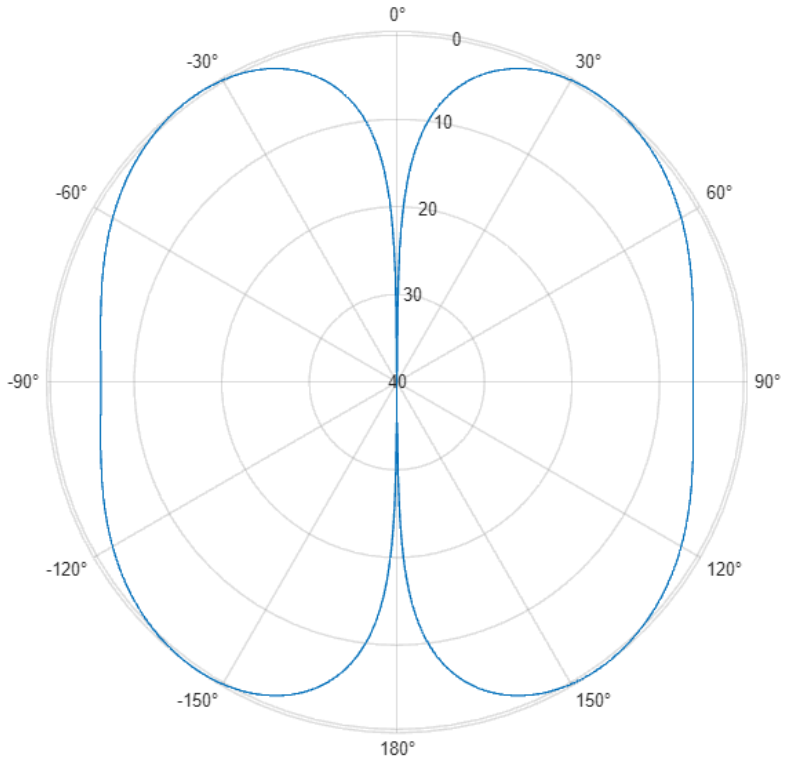

Figure 7: Field Pattern of a loop antenna with radius $a = \lambda/2$.

Appendix A: Spherical Pattern Plot

The figure in Balanis featured a 3D pattern for Question 2. Here is a way to obtain the same plot:

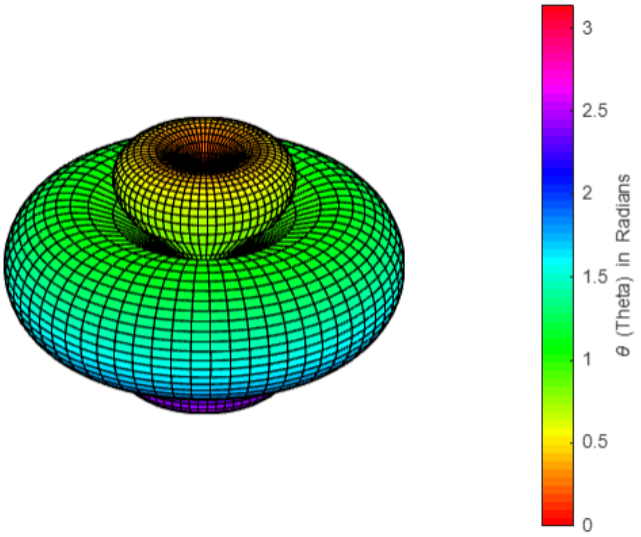

Figure B1: Spherical Pattern of Question 2

Matlab Code for the Spherical Plot:

```
theta = linspace(0, pi, 64);
phi   = linspace(0, 2*pi, 64);

[Phi, Theta] = meshgrid(phi, theta);

E2_1 = (cos(2*pi*1.25*0.5*cos(abs(Theta)))-cos(2*pi*1.25*0.5))./sin(abs(Theta));
E2_1 = abs(E2_1);
E2_1 = E2_1/max(E2_1);
E2_1 = 20*log10(E2_1+10^-12);
E2_1 = (E2_1 + 40).*((E2_1+40)>=0)/40;

r = E2_1;

X = r .* sin(Theta) .* cos(Phi);
Y = r .* sin(Theta) .* sin(Phi);
Z = r .* cos(Theta);

surf(X, Y, Z, Theta);
colormap(hsv(64));
colorbar
cb = colorbar;
cb.Label.String = '\theta (Theta) in Radians';
axis equal
box off
grid off
axis off
```

Appendix B: MATLAB Code

Q1

```
theta_deg_180 = linspace(-180,180,2000);
theta_180 = deg2rad(theta_deg_180);
clf;

%Generate e-fields for Q1
lengths = [1/4, 1/2, 1];
E1_1 =
(cos(2*pi*lengths(1)*0.5*cos(theta_180))-cos(2*pi*lengths(1)*0.5))./sin(theta_180);
E1_2 =
(cos(2*pi*lengths(2)*0.5*cos(theta_180))-cos(2*pi*lengths(2)*0.5))./sin(theta_180);
E1_3 =
(cos(2*pi*lengths(3)*0.5*cos(theta_180))-cos(2*pi*lengths(3)*0.5))./sin(theta_180);

%plot the values
eps = 10^-12;
field_pattern_plot(theta_180,E1_1,'--')
hold on
field_pattern_plot(theta_180,E1_2,'-.')
field_pattern_plot(theta_180,E1_3,'-')
legend({'\lambda/4','\lambda/2','\lambda'},'Location','south')
hold off
```

Q2

```
E2_1 = (cos(2*pi*1.25*0.5*cos(theta_180))-cos(2*pi*1.25*0.5))./sin(theta_180);
field_pattern_plot(theta_180,E2_1,'-')
```

Q3

```
h = [0.25 0.5 1];
theta_deg_90 = linspace(-90,90,1000);
theta_90 = deg2rad(theta_deg_90);
E3_1 = sin(theta_90).*cos(2*pi*h(1)*cos(theta_90));
E3_2 = sin(theta_90).*cos(2*pi*h(2)*cos(theta_90));
E3_3 = sin(theta_90).*cos(2*pi*h(3)*cos(theta_90));

field_pattern_plot(theta_90,E3_1,'--')
hold on
field_pattern_plot(theta_90,E3_2,'-.')
field_pattern_plot(theta_90,E3_3,'-')
legend({'\lambda/4','\lambda/2','\lambda'},'Location','south')
hold off
```

Q4

```
E4_1 = sin(theta_90).*cos(2*pi*5*cos(theta_90));
field_pattern_plot(theta_90,E4_1,'-')
```

Q5

```
h = [0 0.25 0.5 1];

E5_1 = cos(theta_90).^2; %pattern of hertzian dipole by itself
E5_2 = cos(theta_90).*sin(2*pi*h(2)*cos(theta_90));
E5_3 = cos(theta_90).*sin(2*pi*h(3)*cos(theta_90));
E5_4 = cos(theta_90).*sin(2*pi*h(4)*cos(theta_90));

field_pattern_plot(theta_90,E5_1,'--')
hold on
field_pattern_plot(theta_90,E5_2,'-.')
field_pattern_plot(theta_90,E5_3,'-')
field_pattern_plot(theta_90,E5_4,'-')
legend({'0','\lambda/4','\lambda/2','\lambda'},'Location','north')
hold off
```

Q6

```
E6_1 = cos(theta_90).*sin(2*pi*5*cos(theta_90));
field_pattern_plot(theta_90,E6_1,'-')
```

Q7

```
E7_1 = besselj(1,2*pi*0.5*sin(theta_180));
field_pattern_plot(theta_180,E7_1,'-')
```

Functions

```
function y = normalise(x)
    y = x/max(x);
end

function y = trunc(x) % rounds everything below -40dB up to -40dB. Maps -40,0 to 0,1
    y = ((x+40).*(x+40>0)./40);
end

function field_pattern_plot(theta,E1,style)
    eps = 10^-12;
    polarplot(theta,trunc(20*log10(eps+normalise(abs(E1)))),style)
    thetalim([min(theta)*180/pi max(theta)*180/pi]);               % only 0-180°
    rlim([0 1])
    rticks([0 0.25 0.5 0.75 0.99])
    rticklabels([40 30 20 10 0]);         % dB rings

    thetaticks(min(theta)*180/pi:30:max(theta)*180/pi);
    ax = gca;
    ax.ThetaZeroLocation = 'top';    % 0° at the top
    ax.ThetaDir = 'clockwise';       % match antenna books
    set(gcf,'position',[1 , 1 , 700 , 700])

end
```